

Lecture Scribe Marginal PMF



L18.pdf
PDF

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Here is your **exam-ready lecture scribe** based strictly on the provided PDF . I have followed your instruction: **step-by-step reasoning, definitions → derivations → interpretations → examples**, without adding anything outside the given content.



Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1. Marginal PMF

Definition

The key idea:

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

Notation

If joint PMF is $p_{X,Y}(x, y)$, then:

- Marginal PMF of X :

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

- Marginal PMF of Y :

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Marginal PMF from Joint PMF Table

Step-by-step procedure:

1. Start with the **joint PMF table**
 2. To find $p_X(x)$:
 - Fix x
 - Sum across all y
 3. To find $p_Y(y)$:
 - Fix y
 - Sum across all x
-

Example (as per slides)

Given: Joint PMF

Solution Steps:

1. Apply definition:

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

2. Similarly:

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

3. Compute values by summing corresponding rows/columns

Conclusion:

Marginal PMFs are obtained directly by summation over the joint PMF.

2. Correlation

Definition

A basic way to measure joint behavior:

$$R_{XY} = E[XY]$$

Interpretation

- It gives the **average value of product XY**
 - Called a **raw correlation measure**
-

Important Observations

1. Depends on **scale of X and Y**
2. Computed about the **origin (not mean)**
3. Mixes:
 - Mean values
 - Spread
 - Dependence

👉 Therefore:

Not a clean measure of dependence

Special Case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about origin}$$

Transition to Covariance

To remove mean effects:

We center variables → leads to **covariance**

3. Covariance

Definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Step-by-Step Interpretation

1. Subtract mean from each variable
2. Measure how deviations vary together

Meaning:

- Positive → move in same direction
 - Negative → move in opposite direction
-

4. Covariance: Algebraic Form

Start with:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Expand:

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

Apply linearity of expectation:

$$= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y]$$

Simplify:

$$\boxed{\text{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

5. Properties of Covariance

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance as Special Case

$$\text{Var}(X) = \text{Cov}(X, X)$$

3. Linearity with Constants

(Covariance behaves linearly when constants are involved)

4. Independence Property

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

⚠ Important:

Zero covariance $\Rightarrow$ uncorrelated
But NOT necessarily independent

6. Covariance Example 1

Method (as given)

1. Use formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

2. Substitute:

- $E[XY]$
- $E[X]$
- $E[Y]$

3. Simplify step-by-step
 4. Final result obtained after substitution
-

7. Pearson's Correlation Coefficient

Definition

Measures effect of change in one variable when the other changes

Key Points

- Measures strength and direction of linear relationship
 - Normalized version of covariance
 - Unit-free
-

8. Covariance Example 2 (Continuous Case)

Given:

Joint PDF $f_{X,Y}(x, y)$

Steps:

1. Compute:

$$E[X], \quad E[Y], \quad E[XY]$$

2. Use:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

3. Perform integration over valid region
 4. Substitute results
 5. Final covariance obtained
-

9. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is jointly Gaussian if:

- Joint density has **bivariate Gaussian form**
 - Marginals are also Gaussian
-

Interpretation

- Models continuous random variables influenced by many small effects
-

Real-Life Applications

- Signal processing
 - Communication systems
 - Noisy measurements
-

Example (Given in slides)

Climate System Example

- X : CO₂ concentration
- Y : Temperature anomaly

👉 Due to multiple random factors:

Their joint behavior can be approximated using a **Joint Gaussian Distribution**

← END Final Flow Summary (Exam Revision)

1. Marginal PMF

- Sum over other variable

2. Correlation

- $E[XY]$, raw measure

3. Covariance

- Centered version
- $E[XY] - E[X]E[Y]$

4. Properties

- Symmetry, variance, independence

5. Pearson Coefficient

- Normalized, unit-free

6. Joint Gaussian

- Bivariate Gaussian + Gaussian marginals

If you want, I can next:

- Convert this into **1-page revision notes**
- Or solve **expected exam questions step-by-step** from this lecture

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Here is your **strict, exam-focused lecture scribe** based only on the provided PDF .

I have followed your instructions carefully: **definitions** → **notation** → **derivations** →

examples → interpretation, with no extra concepts added.

Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1. Marginal PMF

Definition

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

Notation

Let joint PMF be $p_{X,Y}(x, y)$

- Marginal PMF of X :

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

- Marginal PMF of Y :

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Marginal PMF from Joint PMF Table (Procedure)

Step-by-step:

1. Start with joint PMF table
 2. For $p_X(x)$:
 - Fix x
 - Sum over all y
 3. For $p_Y(y)$:
 - Fix y
 - Sum over all x
-

Example (Given)

Step 1:

Start with joint PMF

Step 2:

Apply definition:

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

Step 3:

Similarly:

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Step 4:

Compute values using summation

Final:

Marginal PMFs obtained

2. Correlation

Definition

$$R_{XY} = E[XY]$$

Interpretation

- Measures **average value of product XY**
 - Called **raw correlation measure**
-

Observations

1. Depends on scale of X and Y
2. Computed about the origin (not means)
3. Mixes:

- Mean
- Spread
- Dependence

👉 Hence:

Not a clean measure of dependence

Special Case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about origin}$$

Transition

To remove mean effect:

| Center variables → leads to covariance

3. Covariance

Definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Interpretation (Step-by-step)

1. Subtract mean from each variable
2. Measure joint variation of centered variables

👉 Meaning:

- Positive → same direction
 - Negative → opposite direction
-

4. Covariance: Algebraic Derivation

Start with:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Step-by-step Expansion

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

Apply expectation:

$$= E[XY] - E[XE[Y]] - E[YE[X]] + E[E[X]E[Y]]$$

Simplify

Since constants:

$$E[XE[Y]] = E[X]E[Y]$$

$$E[YE[X]] = E[Y]E[X]$$

Thus:

$$= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y]$$

Final Result

$$\boxed{\text{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

5. Covariance Properties

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance Special Case

$$\text{Var}(X) = \text{Cov}(X, X)$$

3. Linearity w.r.t Constants

Covariance behaves linearly with constants

4. Independence

$$X \text{ independent of } Y \Rightarrow \text{Cov}(X, Y) = 0$$

Important:

Zero covariance $\Rightarrow$ uncorrelated
Does NOT imply independence

6. Covariance Example 1

Given Hint

Substitute values into covariance formula

Step-by-step Solution

Step 1:

Use formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2:

Substitute:

- $E[XY]$
 - $E[X]$
 - $E[Y]$
-

Step 3:

Compute expression

Step 4:

Since given expressions are substituted:

Simplify step-by-step

Final:

Covariance obtained

7. Pearson's Correlation Coefficient

Definition

Measures the effect of change in one variable when the other variable changes

Properties

- Measures strength and direction of linear relationship
 - Normalized covariance
 - Unit-free
-

8. Covariance Example 2 (Continuous Case)

Given

Joint PDF $f_{X,Y}(x, y)$

Step-by-step Solution

Step 1:

Compute:

$$E[X], \quad E[Y], \quad E[XY]$$

Step 2:

Use covariance formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 3:

Evaluate integrals over valid region

Step 4:

Substitute computed values

Step 5:

Simplify

Final:

Covariance obtained

9. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is jointly Gaussian if:

- Joint density is bivariate Gaussian
 - Marginals are Gaussian
-

Interpretation

- Models continuous random variables influenced by many small random effects
-

Applications (Given)

- Signal processing
 - Communication systems
 - Noisy measurements
-

Example (Given)

Climate System

- X : CO₂ concentration
- Y : Temperature anomaly

Explanation:

Due to many random processes:

Their joint behavior can be approximated by a joint Gaussian distribution

Final Exam Flow

1. Marginal PMF → summation
2. Correlation → $E[XY]$
3. Covariance → centered measure
4. Algebraic form → $E[XY] - E[X]E[Y]$
5. Properties → symmetry, independence
6. Pearson → normalized covariance
7. Joint Gaussian → bivariate normal

If you want next:

- I can convert this into **very short revision notes (1–2 pages)**
- Or give **important exam questions from this lecture with full solutions**

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understand easily. Do not include any points or derivations which is not present in the context.

Here is your **fully structured, step-by-step lecture scribe** strictly based on the provided PDF .

I have carefully followed your instructions: **clear logical flow → definitions → notation → derivations → examples exactly as given → no extra content added.**

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1. Marginal PMF

Definition

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

Notation

Let joint PMF be $p_{X,Y}(x, y)$

- Marginal PMF of X :

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

- Marginal PMF of Y :

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Marginal PMF from Joint PMF Table (Step-by-step)

1. Start with joint PMF table
2. To compute $p_X(x)$:
 - Fix x
 - Sum probabilities across all y

3. To compute $p_Y(y)$:

- Fix y
 - Sum probabilities across all x
-

Example (Given)

Step 1:

Joint PMF is given

Step 2:

Apply definition:

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

Step 3:

Similarly:

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Step 4:

Perform summation across rows/columns

Final Result:

Marginal PMFs obtained

2. Correlation

Definition

$$R_{XY} = E[XY]$$

Interpretation

- Measures average value of product XY
- Called raw correlation measure

Important Observations

1. Depends on scale of X and Y
2. Computed about the origin (not about means)
3. Mixes:
 - Mean
 - Spread
 - Dependence

Special Case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

Transition to Covariance

To remove effect of means, we center variables → leads to covariance

3. Covariance

Definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Interpretation (Step-by-step)

1. Subtract mean from each variable
2. Measure how centered variables vary together

Meaning

- Positive covariance → deviations occur in same direction
 - Negative covariance → deviations occur in opposite direction
-

4. Covariance: Algebraic Form

Start with Definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Step-by-step Derivation

Step 1: Expand product

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

Step 2: Apply expectation

$$= E[XY] - E[XE[Y]] - E[YE[X]] + E[E[X]E[Y]]$$

Step 3: Use constant property

$$E[XE[Y]] = E[X]E[Y]$$

$$E[YE[X]] = E[Y]E[X]$$

Step 4: Substitute

$$= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y]$$

Final Result

$$\boxed{\text{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

5. Covariance Properties

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance Special Case

$$\text{Var}(X) = \text{Cov}(X, X)$$

3. Linearity with Respect to Constants

Covariance is linear when constants are involved

4. Independence

$$X \text{ independent of } Y \Rightarrow \text{Cov}(X, Y) = 0$$

Important Note

Zero covariance means uncorrelated, but does not imply independence

6. Covariance Example 1

Given Hint

Substitute both expressions into the covariance formula

Step-by-step Solution

Step 1:

Use formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2:

Substitute given expressions for:

- $E[XY]$
 - $E[X]$
 - $E[Y]$
-

Step 3:

Substitute values into covariance formula

Step 4:

Since expressions are known, simplify

Final Result:

Covariance obtained after simplification

7. Pearson's Correlation Coefficient

Definition

Pearson's correlation coefficient calculates the effect of change in one variable when the other variable changes

Properties

- Measures strength and direction of linear relationship
 - Normalized version of covariance
 - Unit-free
-

8. Covariance Example 2

Given

Joint probability density function $f_{X,Y}(x,y)$

Step-by-step Solution

Step 1:

Compute expectations:

$$E[X], \quad E[Y], \quad E[XY]$$

Step 2:

Use formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 3:

Evaluate integrals over given region

Step 4:

Substitute values

Step 5:

Simplify

Final Result:

Covariance obtained

9. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is jointly Gaussian if:

- Joint density has bivariate Gaussian form
 - Marginals are Gaussian
-

Interpretation

- Used to model continuous variables influenced by many random effects
-

Applications (Given)

- Signal processing
 - Communication systems
 - Noisy measurements
-

Example (Given in Slides)

Climate System

- X : CO₂ concentration
 - Y : Temperature anomaly
-

Explanation

Due to many environmental random processes:

Their joint behavior can be approximated by a Joint Gaussian Distribution

Final Summary Flow (For Exam)

1. Marginal PMF $\rightarrow$ summation over joint PMF
2. Correlation $\rightarrow E[XY]$, raw measure
3. Covariance $\rightarrow$ centered measure
4. Algebraic form $\rightarrow E[XY] - E[X]E[Y]$
5. Properties $\rightarrow$ symmetry, independence
6. Pearson coefficient $\rightarrow$ normalized covariance

7. Joint Gaussian → bivariate Gaussian model

If you want next, I can:

- Make **1-page ultra-short revision sheet**
- Or solve **expected exam numericals step-by-step from this lecture**